

A new normalization method for CyTOF based experiment.

As a reminder what we currently do is assume that the cell permeability multiplicatively changes the markers and thus we find a linear combination of the core histones that minimized the sum of the variances for the core histones when used to divide the markers.

This works, but...

There are several issues with this:

- We implicitly assume that all core histones are affected the same way by the permeability
- We assume all the of variability in the core histones comes from technical effects.
- We assume all markers are affected in the same way by the multiplicative factors.

Based in the scTransform method for scRNA (and scCUT&Tag) data, I want to propose a new normalization scheme, outlines below.

We start with a multiplicative model, defined in the following way:

$$X_{i,m} = B_{i,m} \cdot T_i^{\gamma_m} \cdot \varepsilon_{i,m}$$

Where $X_{i,m}$ is the measured signal, $B_{i,m}$ is the true biology, T_i is a multiplicative factor related to the cell permeability, γ_m is a marker dependent exponent (such that for markers that are unaffected by the permeability $\gamma \rightarrow 0$, and ε is a noise term, which we assume normally distributed around 1.

In order to find T_i we find the first PC for the core histones (and potentially DNA if we have it), since that typically captures the variability related to the permeability and cell size.

Taking the arcsinh transform (as we do for CyTOF data) we get:

$$Y_{i,m} \simeq \alpha_m + \gamma_m \log(T_i) + Bio_{i,m}$$

Thus, **each marker's dependence on the technical factor is linear in arcsinh space**, and γ_m is simply the regression slope.

We now perform a linear regression for each marker to the form:

$$Y_{i,m} = \alpha_m + \gamma_m \log(T_i) + \varepsilon_{i,m}$$

The coefficient γ_m measures the **fraction of marker variance attributable to permeabilization**, not the total biological signal. Markers strongly affected by nuclear staining (e.g., core histones) show $\gamma_m \approx 0.4-0.7$, PTMs show $\gamma_m \approx 0.1-0.5$, and markers whose signal is primarily biological (e.g., Ki-67, H2AK119ub) show $\gamma_m \approx 0$.

Importantly, γ_m **is not expected to equal 1**, even for core histones, because biological heterogeneity (cell cycle phase, chromatin organization, apoptosis) contributes substantially to their variance. γ_m reflects only the *technical* component of variation shared with PC1.

Finally we remove the technical component using:

$$Y_{i,m}^{Corr} = Y_{i,m} - \gamma_m (\ln T_i - \text{Med}(\ln T_i))$$

Centering at the median ensures that **cells with typical permeabilization remain unchanged**, while over- or under-permeabilized cells are adjusted relative to a biological reference state. This anchoring avoids introducing unwanted global shifts in marker expression.

The correction is **marker-specific**, **proportional**, and **conservative**:

- markers with large γ_m receive stronger correction,
- markers with small γ_m are minimally altered,
- markers with $\gamma_m \approx 0$ are effectively unchanged.

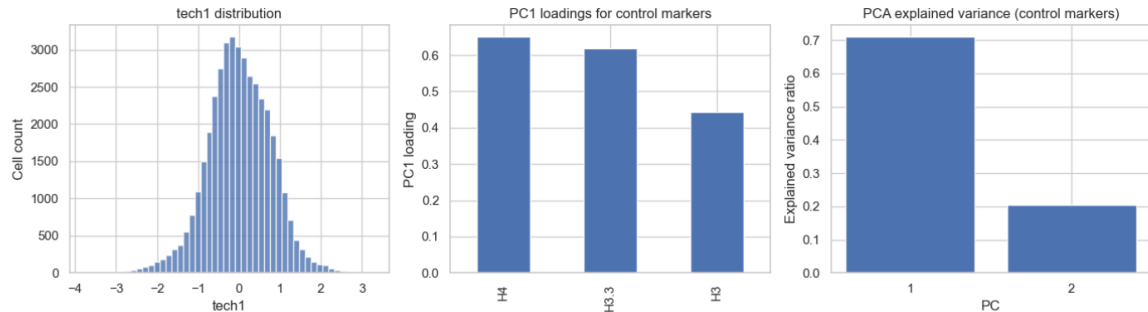
This mimics the per-gene normalization from scTransform).

After the normalization we continue as usual with z-scoring datasets together for comparison. An additional option is to first find clusters based on cell identity markers (in PC space for those markers) and perform per-cluster normalization if we believe there are strong biological effects (for example cell-cycle related effects).

Work flow demonstration:

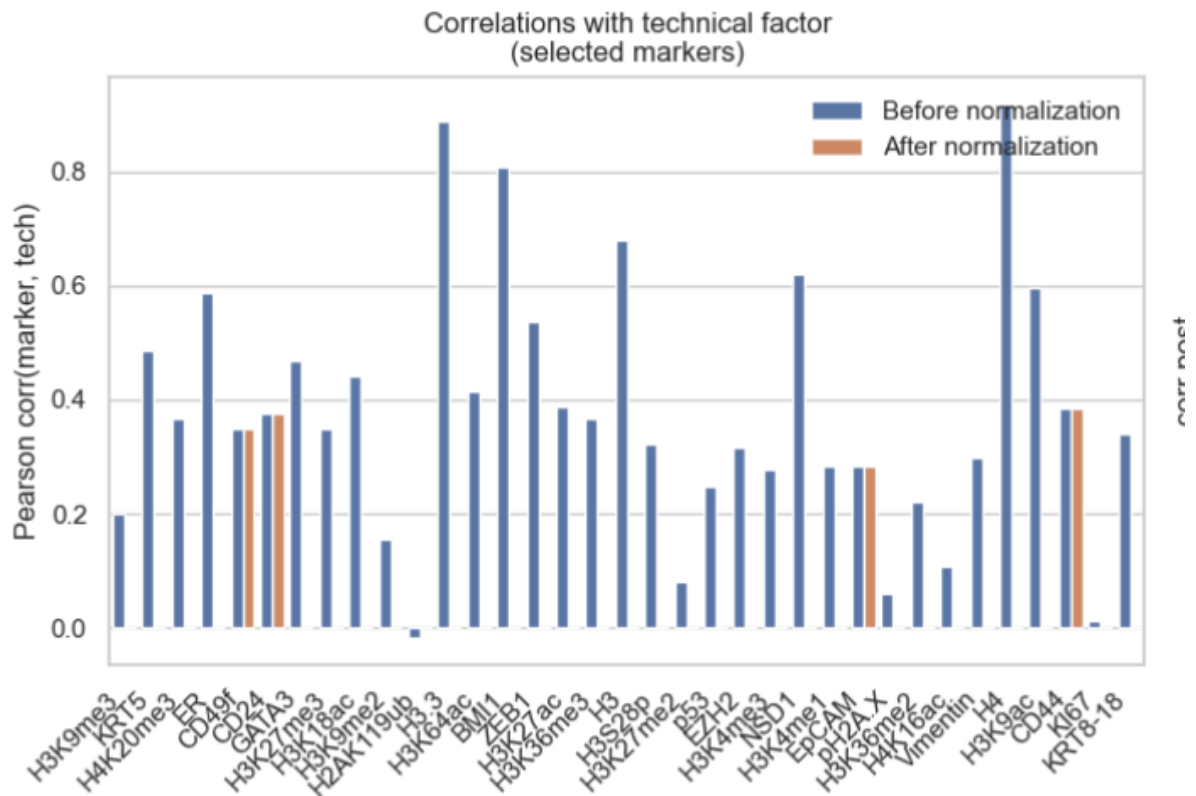
I test on one of the MCF7 clones (clone 5 from the recent experiment.

First I plot for the core histones and the technical factor:



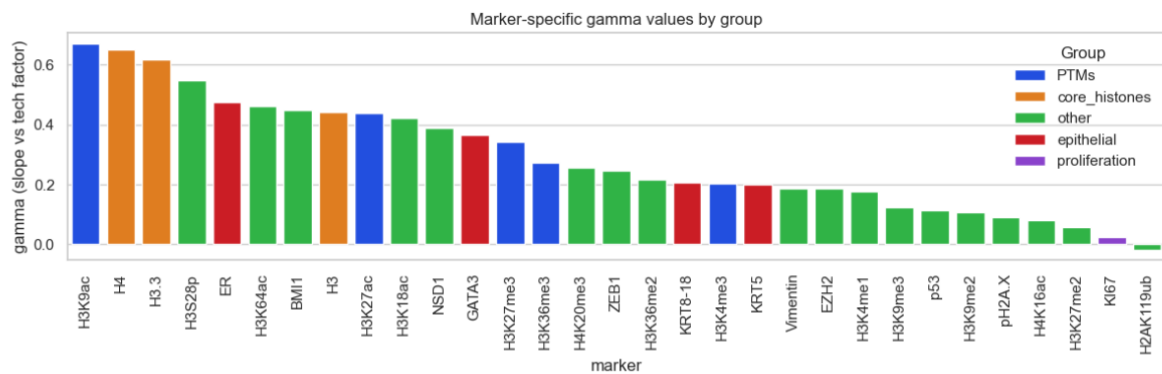
All core histones load in the same direction on PC1, which also explains most of the variance. The distribution of the technical factor over all cells also makes sense, it that it is normally distributed.

I then preform the normalization and test the correlations with the technical factor before and after:



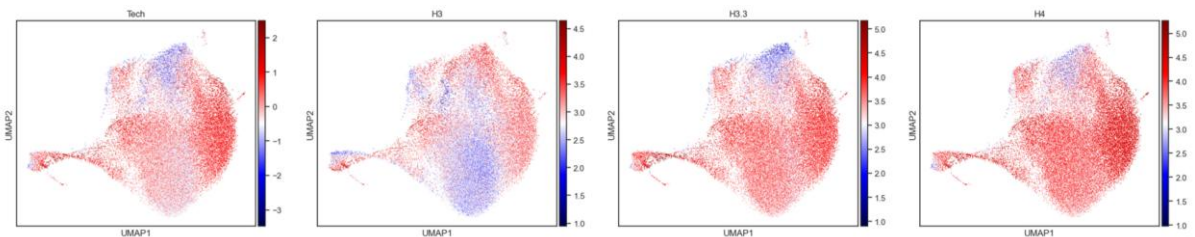
As expected, the correlation of the intra cellular markers with the technical factor drop to zero, while those of the extracellular markers, remain unchanged.

I also plot the distribution of the marker specific exponents:

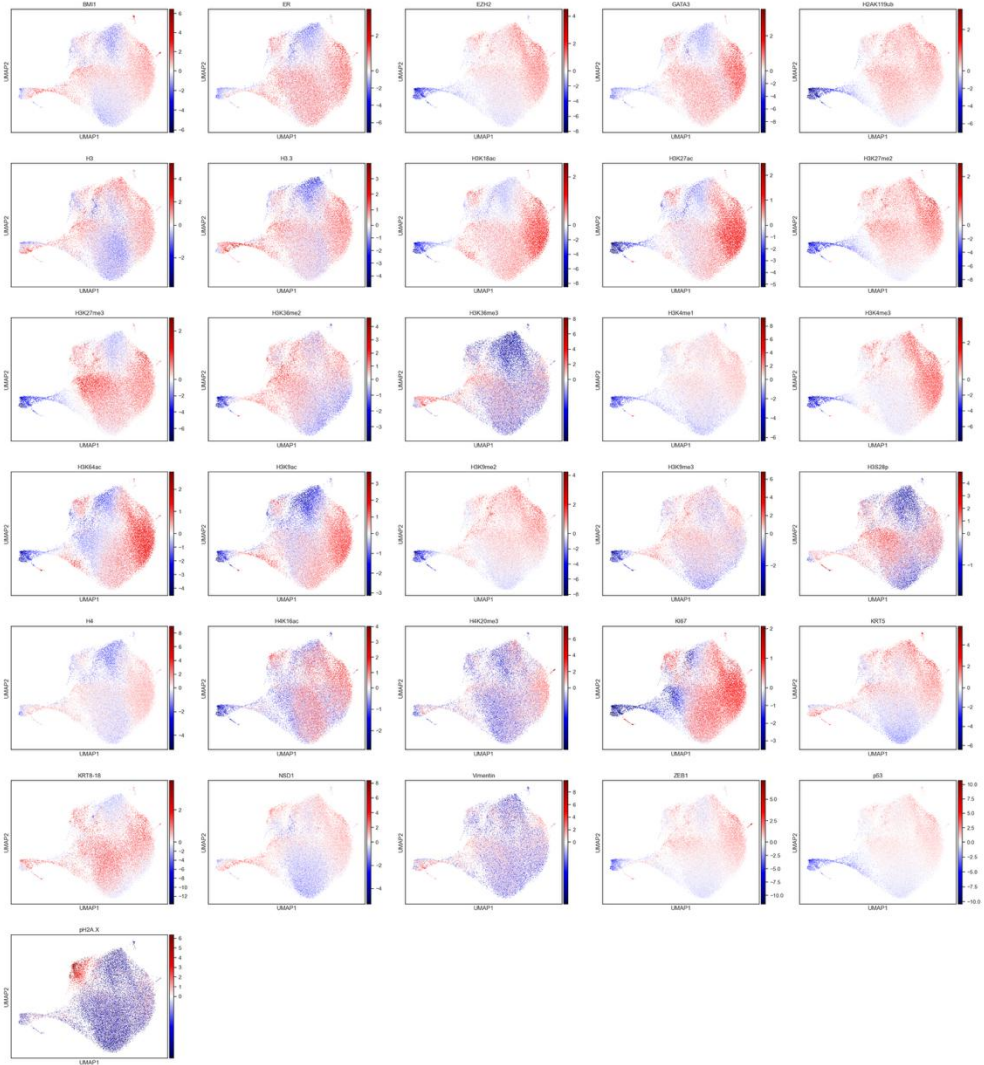


Which also seems to make sense some what.

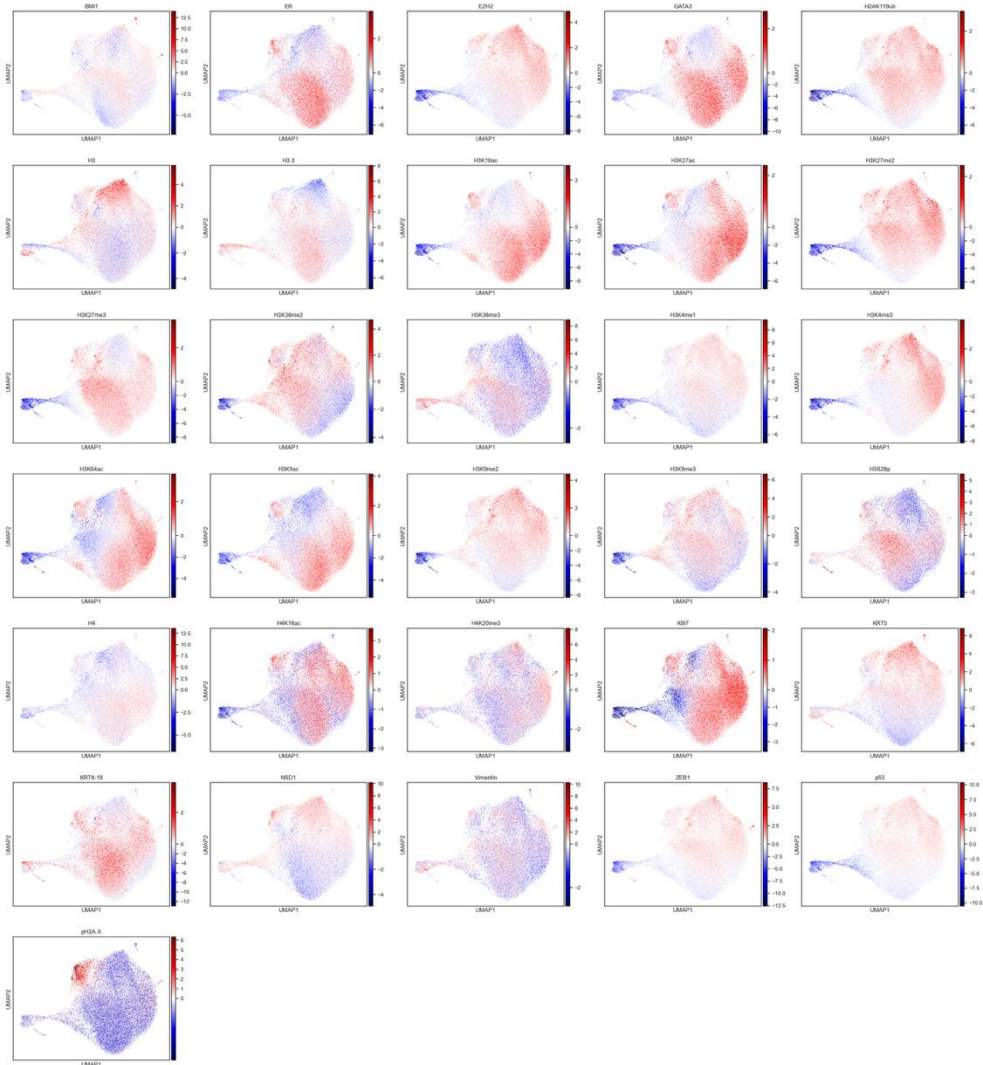
A plot of the technical factor and the core histones for the uncorrected dataset shows that it that tech factor deos indeed follow the core histones:



A UMAP plot of the uncorrected data



And the corrected data:



While the results are not super different from the way we do the normalization I think this method is more biologically motivated and probably a better way to go (but only for new experiments, I would not want to redo everything we did).

